Accessibility, Inclusion, and Usability

What is Accessibility?

Have you ever experienced a product or website that you could not figure out or maybe you purchased a piece of furniture that required you have your own tools and was missing instructions?

Accessibility is defined as a "word used to describe whether a product can be used by people of all abilities.“

Digital accessibility includes web accessibility but also refers to the accessibility of anything digital such as video, audio, electronic documents, animations, kiosks, and mobile apps.

Web accessibility means that websites, tools, and technologies are designed and developed so that people with disabilities can use them.

An easy way to make a website more accessible is to have it available for different languages. It’s easy to just make it English, but then all non-English speaking people will have a different interaction with that site.

Who benefits from Accessibility?

Web Accessibility benefits all users – people with or without disabilities.

Some examples include:

- People with disabilities (low vision, colour blind, low motor control)
- Low literacy or not fluent in the language
- People with low bandwidth connections or using older technologies
- New and infrequent users (computer literacy)
- Older and younger people
- Mobile device users

The last example may surprise you, however at the time of this module’s creation, mobile holds 55% of the marketplace share worldwide. (**https://gs.statcounter.com/platform-market-share/desktop-mobile/worldwide) (https://gs.statcounter.com/platform-market-share/desktop-mobile/worldwide))**

This means that websites need to be designed to work well on mobile phones and desktops (this is referred to as responsive design **https://www.w3schools.com/html/html_responsive.asp) (https://www.w3schools.com/html/html_responsive.asp))**

Remember back to the first section where the module described a scenario where a person was trying to access a store but the only way to enter was from a stairway. Having either a ramp or an elevator/lift to help access the building doesn’t just benefit the person in a wheelchair, but a

product supplier with a dolly of heavy box that they are trying to deliver, or the elderly person who can use stairs, but the ramp is much easier for them.

Designing with accessibility in mind helps everyone.

Inclusion

While accessibility aims to minimize disability in websites, it does not try to address broader issues. This is where inclusion comes in. Inclusive design improves features for everyone.

A great visual explanation of inclusive design can be seen here: **https://www.youtube.com/watch?v=S0EL0nKP314 (https://www.youtube.com/watch?v=S0EL0nKP314)**

Inclusion is a method and not a process for meeting all accessibility standards. It is about diversity and consideration of everyone including people with disability are considered. Internationalization, address inclusion issues.

Usability

Usability is a method that measures how useable a website is.
A good website is measured by it's

- Effectiveness
- Efficiency
- Satisfaction

You will learn more about these points later in the module, but a highly usable website is intuitive and easy to use.

Here are ten examples of bad website/app designs:
(https://careerfoundry.com/en/blog/ux-design/10-classic-ux-design-fails/)

- **https://careerfoundry.com/en/blog/ux-design/10-classic-ux-design-fails/ (https://careerfoundry.com/en/blog/ux-design/10-classic-ux-design-fails/)**

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Defining Disability

Before you start, take a minute to write down what you think “Disability” means? After reading this section how did your answer compare to what you have read?

Disability

The World Health Organization defines disability as "a mismatch in interaction between the features of a person's body and the features of the environment in which they live”.

An example of this is a person in a wheelchair and a set of stairs. The person can move around from one place to another, they can eat sleep just like everyone else, however a certain building in that town has the only entrance to that building accessible by stairs. Now, this creates a disability to enter that building.

Stairs are considered a “normal” design since it does not affect the majority of people, but it is in fact creating a disability.

Disability is NOT a personal health condition but a mismatch of human interactions.

There is a risk of creating a design based only on your own experiences and abilities. This will generate solutions that work well for people like yourselves but could exclude many. The bottom line is that you could, in effect, be creating disability.

Watch the following video for a social model animation on non-inclusive design: **https://www.youtube.com/watch?v=9s3NZaLhcc4 (https://www.youtube.com/watch?v=9s3NZaLhcc4)**

How society views disabilities

“Leaders in the disability rights movement have constructed two distinct models of how society views disabilities: the Medical Model and the Social Model.” (Medical and Social Models of Disability)

	Medical Model (outdated)	Social Model
Disabled	= Less Able	= Disabled by social barriers
Responsibility	Person with an impairment to overcoming disabled barriers	Governments and everyone to build more equal and inclusive society
Focus	prevention, treatment or curing of the disabling condition	Ways to provide accessibility, enables people rather than disables them.

Medical Model

The medical model describes disability as a consequence of a health condition, disease or caused by a trauma that can disrupt the functioning of a person in a physiological or cognitive way. This model is a conceptualization of disability as a condition a person has and focuses on the prevention, treatment or curing of the disabling condition.

According to the medical model, the word disabled means Less able. Less able to achieve your potential. This outdated view of the world puts the responsibility of overcoming disabled barriers on the shoulders of the person with an impairment.

Social Model

This model focuses on barriers facing people with disabilities instead of concentrating on impairments and deficits of the person with a disability. In this model a person's activities are limited not by impairment or condition but by environment and barriers are consequences of a lack of social organization.

The social modal, says that a person doesn't have a disability, but they are disabled by society that prevents them from achieving their potential. The social model was developed by disabled people and their allies to help them act against discrimination and to empower people to find solutions, remove barriers and campaign together for equality and human rights.

The social model looks for ways that society can be planned and organized in order to provide accessibility, independence and opportunity in a way that enables people rather than disables them.

Measuring Disability in Canada

In 2017, 6.2 million (22%) Canadians aged 15 and older had a disability. This is based on a social model of disability. That is more than 1 out of 5 people!

By Gender:

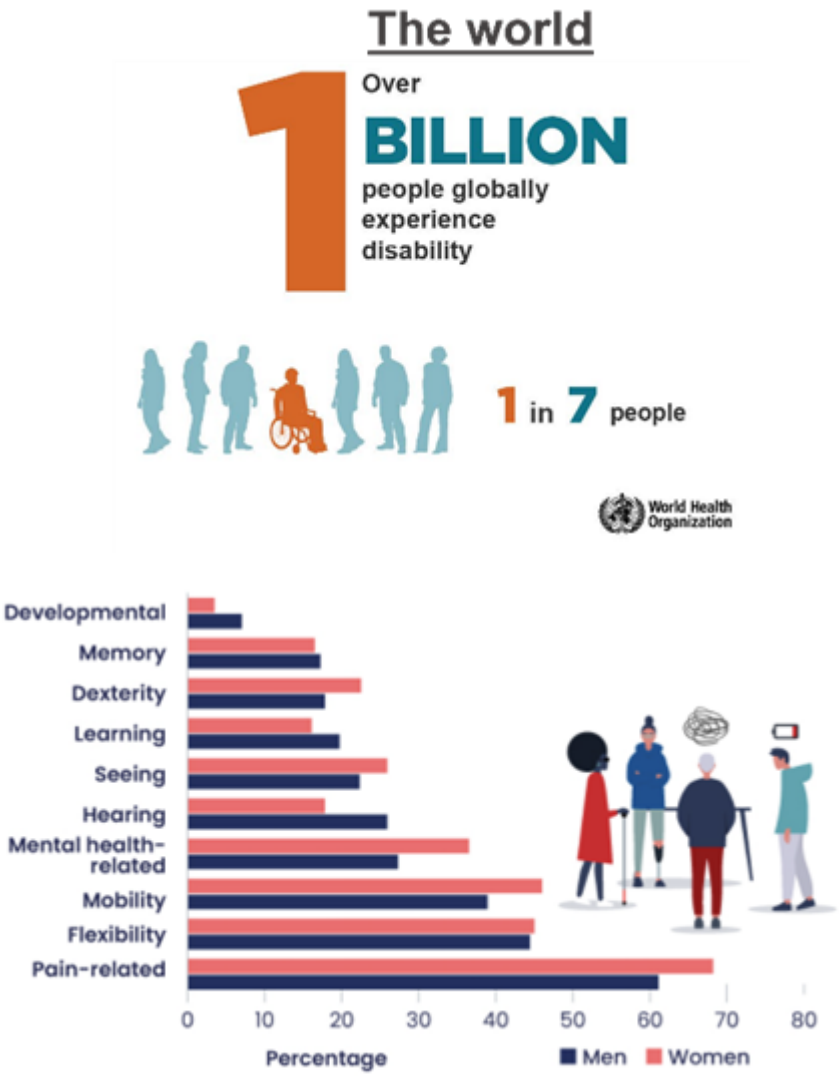


By Age:



By Age group

The World:



Most people have two or more types of disabilities (29% one disability type, 71% two or more types of disabilities)

Most people have two or more types of disabilities (29% one disability type, 71% two or more types of disabilities)

The 2017 CSD (**Canadian Survey on Disability (https://www150.statcan.gc.ca/n1/pub/11-627-m/11-627-m2022062-eng.htm)**) collected data on experiences of persons with disabilities is based on a social model of disabilities, which considers disability to be the result of the interaction between a person's functional limitations and barriers in the environment, including social and physical barriers that make it harder to function day to day.

Now that you have learned what disabilities are it is important to note that the social model means that the more barriers get removed, the less percentages of disabilities occur.

Reflection

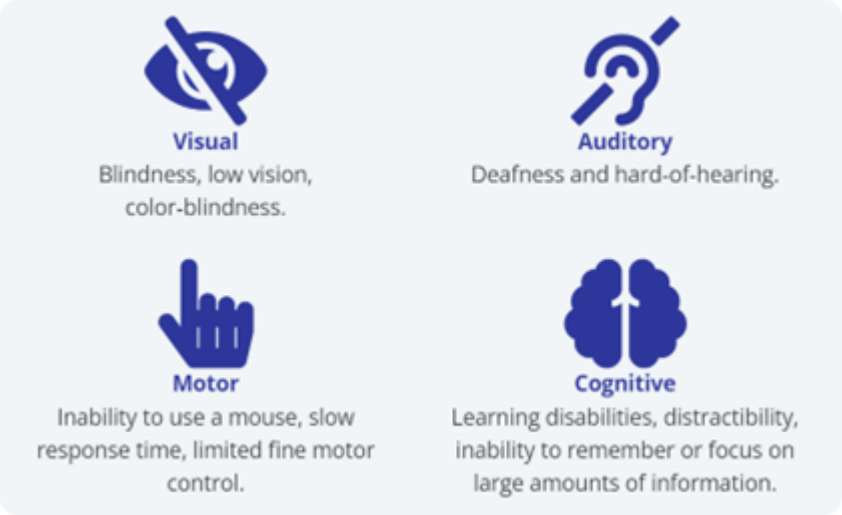
How did your description you wrote down differ from the description in the module?

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Diversity of Users and Accessibility Barriers

Not all people with medical disabilities encounter barriers on the Web, and those with different types of disabilities encounter different types of barriers. For instance, if a person is in a wheelchair, they may encounter no barriers at all in web content. A person who is blind will experience different barriers than a person with limited vision. Different types of disabilities and some of their commonly associated barriers are described in this section;

Categories of Disabilities

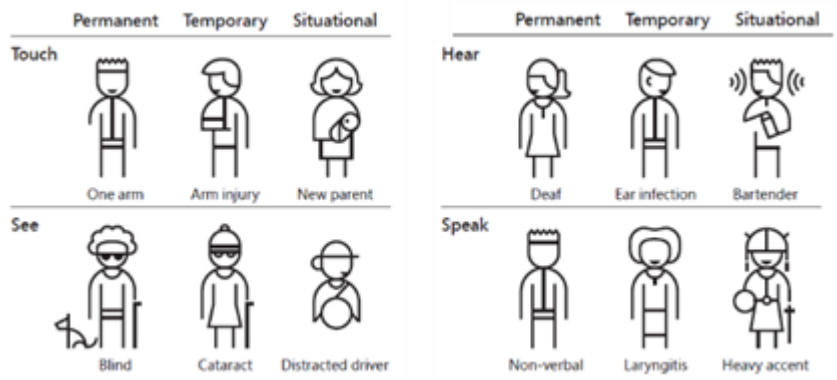


There are four main disabilities barriers that people may encounter on the web. Visual, Auditory, Motor, and Cognitive. Cognitive disabilities are also known as invisible disabilities.

Types of Disabilities

Everyone has different levels of abilities and limits to those abilities. Your abilities also change based off of your current situation, when you are stressed, you may lack concentration. If you are fine at typing and using a mouse, then you may have problems with that same task if you broke your arm.
A solution that can help remove disability barriers for one person can help others as well.

Below is the Microsoft Persona Spectrum that describes different types and categories of disability.



Experience Dyslexia

There is a video that will play a wall of text. Your job is to read the text in two minutes.

- **https://youtu.be/69OWA6mFII4 (https://youtu.be/69OWA6mFII4)**

Afterwards, ask yourself the following questions below.

Questions to discuss after the experiment:

- Were you able to complete the reading?
- Could you answer the questions?
- How did you feel during the activity?
- How to create a dyslexia friendly web content?

Original text used in the dyslexia experiment.
https://education.nationalgeographic.org/resource/beyond-king-tut/ (https://education.nationalgeographic.org/resource/beyond-king-tut/)

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Essential Components of Web Accessibility

In this section you will learn about the guidelines and federal & provincial legislations for Information and Communications Technology (ICT).

Web accessibility depends on the cooperation of seven components:

- Content (text, images, sounds, code, markup, presentation)
- User agents (web browsers, media players)
- Assistive technologies (screen readers, keyboards, switches)
- Users' knowledge and experiences (user behaviours)
- Developers (designers, coders, authors)
- Authoring tools (software that creates websites, content)
- Evaluation tools (HTML validators, CSS validators)

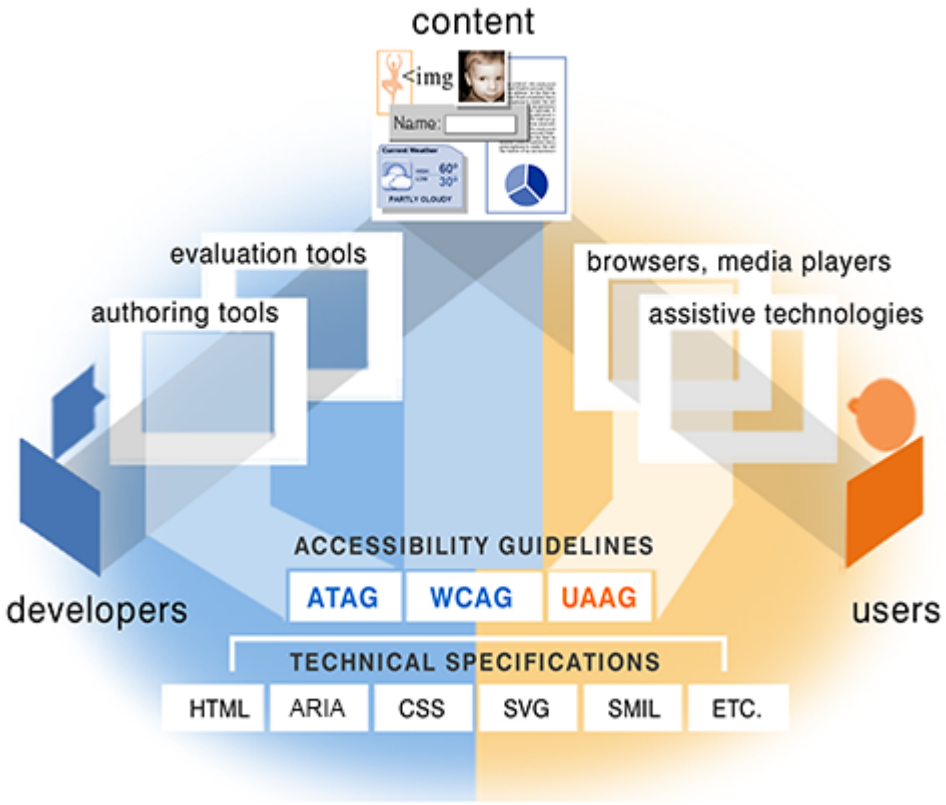


Fig 1.8

In order for this image to be accessible online there is a few steps to get it to that point.

- An accessible image must have an alt attribute (this is part of the HTML "img" tag. It is used for text to speech programs and to display if the image itself cannot load"

- To have an image on a web page, the developer uses a software, Dreamweaver or word press or team site to add the image
- The software supports the alt attribute and gives the option with a right click on the image to enter the text of the alt
- Then the developer makes validates the code with a tool like the W3C validator tool (**https://validator.w3.org/) (https://validator.w3.org/))**
- The W3c Validator tool supports the alt attribute and verifies whether the alt attribute exists on the image tag
- The page is online for users to access it
- The user uses chrome(or another browser) to see the page with the image
- The browser supports the alt attribute and passes the alt to the user and API accessibility for JAWS (Job Access with Speech, a screen reading tool)
- A blind man uses JAWS
- Jaws recognizes the alt and reads it for the user
- W3C puts the rules on accessibility for each component
  So when an accessibility feature is supported by all components you will have an accessible product.

But if one of the components of the support is not present, it will be demotivating for others to use it and by consequence creates less accessible web products.

Img src and more information on web accessibility components can be found here:
https://w3.org/WAI/fundamentals/components/
(https://w3.org/WAI/fundamentals/components/)
Image by Michael Duffy, from: Essential Components of Web Accessibility. S.L. Henry, ed.
Copyright W3C ® (MIT, ERCIM, Keio, Beihang). w3.org/WAI/fundamentals/components/

WCAG, ATAG, UAAG, and WAI-ARIA are all accessibility standards and guidelines developed by the World Wide Web Consortium's (W3C) Web Accessibility Initiative (WAI). While they all share a common goal of making the web more accessible to people with disabilities, they have different focuses and objectives.

Web Content Accessibility Guidelines (WCAG)

WCAG focuses on making web content accessible to people with disabilities, such as visual, auditory, and cognitive impairments. WCAG provides a set of technical standards that define how to make web content perceivable, operable, understandable, and robust. It is organized into three levels of conformance, with Level A being the minimum level and Level AAA being the highest.

W3C Authoring Tool Accessibility Guidelines (ATAG)

ATAG focuses on making authoring tools accessible, such as content management systems and web design software, more accessible to content creators with disabilities. The guidelines cover the entire authoring process, from creating, editing, and publishing web content to ensuring that the content is accessible to people with disabilities.

User Agent Accessibility Guidelines (UAAG)

UAAG (User Agent Accessibility Guidelines) provides guidelines for making web user agents, such as web browsers and media players, more accessible to people with disabilities. The guidelines cover a range of features, such as keyboard navigation, text resizing, and audio descriptions, that make it easier for people with disabilities to use web content.

WAI's Accessible Rich Internet Applications Guidelines (WAI-ARIA)

WAI-ARIA provides technical specifications for making complex web applications accessible to assistive technologies.

In summary, WCAG focuses on making web content accessible, ATAG focuses on making authoring tools accessible, UAAG focuses on making web user agents accessible, and WAI-ARIA provides technical specifications for making complex web applications accessible to assistive technologies.

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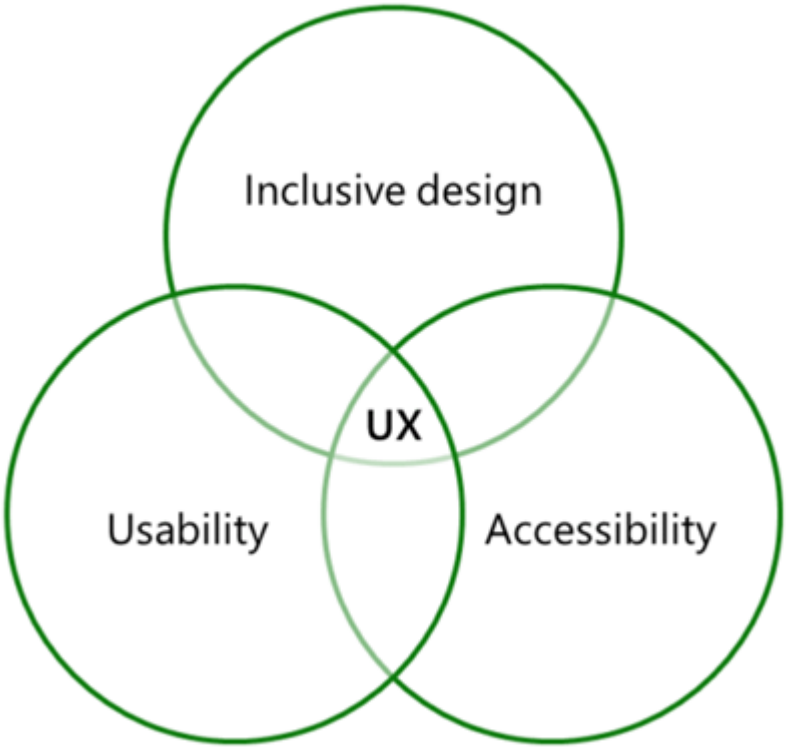
Improving User Experience

From what you have read in the previous sections you have learned that websites can be improved upon to help remove disability barriers and improve quality of life for everyone.

Using Accessibility, Inclusion and Usability these all improve the User's Experience (UX)

Areas to distinguish

- Accessibility focuses on disability and does not try to address broader issues.
- Inclusion includes the broad range of issues such as computer literacy, economic situation, geographic location, culture. Internationalization, address inclusion issues.
- Usability means that a website is intuitive and easy to use.



Consideration of inclusive design, usability and accessibility makes all IT solutions and equipment usable by all.

Remember, Usable by All is what the overarching goal of the Accessible Canada Act (check related slide).

An example of good user experience is that you include diversity of people in the design phase of the product, test it with them during and after development. Design with users, iterate and improve in response to user needs, build in accessibility from the start.

Watch the following video: **https://www.w3.org/WAI/perspective-videos/layout/ (https://www.w3.org/WAI/perspective-videos/layout/)**

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Legal Accessibility Requirements

Accessibility for Ontarians with Disabilities Act (AODA)

AODA aims to make Ontario fully accessible for people with disabilities by 2025 through the use of its information and communications standard.

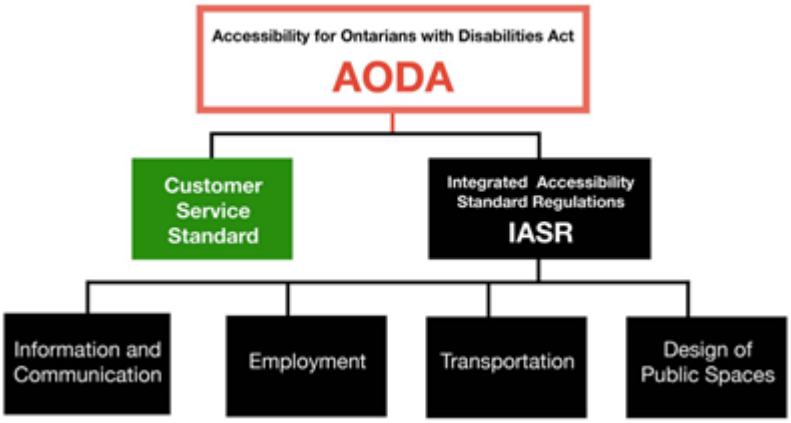
Information and Communications Standard

- Ensures that all information and communications* provided by organizations in Ontario are accessible to people with disabilities.
- By January 1, 2025: All public websites and web content must conform to WCAG 2.0 Level AA, including live captioning and pre-recorded audio descriptions.
- Compliance with the WCAG 2.0 Level AA is a legal requirement under the AODA and failure to comply can result in penalties and fines.

* Information and communications include websites, public information, and communications with customers and employees.

There are five AODA standards. The Customer Service Standard was the first standard to become law in 2008.

The information and communications, employment, transportation, and design of public spaces standards have been combined under one regulation called the Integrated Accessibility Standards Regulation (IASR). (See Fig.1.9)



These five standards are now law, and the requirements are being phased in over time. The phased-in approach gives businesses and organizations time to make accessibility a part of their daily business.
(https://www.ontario.ca/page/how-make-websites-accessible)

- **https://www.ontario.ca/page/how-make-websites-accessible (https://www.ontario.ca/page/how-make-websites-accessible)**

Accessible Canada Act (ACA)

The ACA came into force in 2019. The overarching goal of ACA is to realize a barrier-free Canada by 2040.

The Government of Canada Information and Communication technologies (ICT) accessibility strategy is as follows.

Current state (not sufficient to meet ACA)	Desired state
Standard on Web Accessibility WCAG 2.0 AA	Harmonised European Standard, EN 301 549 (2018) WCAG 2.1 AA
• Cover only externally public facing websites • Target government of Canada clients • Does not cover other aspects of ICT	• Cover external and internal • Target clients and employees • Cover all ICT *

* Examples of ICT are web pages, electronic content, telecommunications products, computers and ancillary equipment, software including mobile applications, information kiosks and transaction machines, videos, IT services, and multifunction office machines which copy, scan, and fax document

The ACA applies to a wide range of federally regulated entities. This includes, for example:

- industry sectors such as banking, telecommunications, and transportation
- Government of Canada departments and agencies
- Parliament
- Crown corporations
- Canadian Armed Forces
- Royal Canadian Mounted Police, and
- First Nations band councils

The ACA requires regulated entities to prepare and publish accessibility plans, set up feedback processes and report openly on progress. The ACA gives the authority to create and enforce accessibility regulations.

With the ACA coming into force in 2019 , the Accessibility Strategy was launched in 2019 as well to help the public service lead by example in accessibility and to prepare organizations for the Accessible Canada Act. The Accessibility Strategy has 5 priority areas: Employment; Built Environment; Technology; Services; and Culture.

The ACA's 7 priority areas are shown in Fig. 1.11



Continued readings:
(https://www.tbs-sct.canada.ca/pol/doc-eng.aspx?id=32620&section=html)

https://www.tbs-sct.canada.ca/pol/doc-eng.aspx?id=32620&section=html (https://www.tbs-sct.canada.ca/pol/doc-eng.aspx?id=32620&section=html)
https://www.canada.ca/en/government/publicservice/wellness-inclusion-diversity-public-service/diversity-inclusion-public-service/accessibility-public-service/accessibility-strategy-public-service-toc/accessibility-strategy-public-service-technology.html (https://www.canada.ca/en/government/publicservice/wellness-inclusion-diversity-public-service/diversity-inclusion-public-service/accessibility-public-service/accessibility-strategy-public-service-toc/accessibility-strategy-public-service-technology.html)

Quick summary of the ACA:

- https://www.youtube.com/watch?v=daP_7MjpDCs (https://www.youtube.com/watch?v=daP_7MjpDCs)

EN 301 549: European standard for digital accessibility

EN 301 549 is the European standard of Accessibility requirements for ICT products and services in the European Union.

It adopts the latest version of the W3C's Web Content Accessibility Guidelines (WCAG), in this case, it directly references WCAG 2.1 AA.

Which includes requirements for the procurement of accessible ICT products and services by public sector organizations in Europe.

Many governments guidelines including Government of Canada encourage alignment with EN 301 549.

If you would like to learn more view this video on the EN 301 549 standard: **https://www.youtube.com/watch?v=J-BEi2yyuFw (https://www.youtube.com/watch?v=J-BEi2yyuFw)**

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Things in life are not always designed with everyone in mind. This can create disabilities for different people who are trying to use them. In this module you will learn what a disability, accessibility and inclusion are. What the accessibility requirements are for the Information and Communication Technology field, and why the need for accessible design is important.

Learning Objectives

By the end of this module, you should be able to complete the following:

01 Differentiate between Social and Medical Models of Disabilities

02 Determine if a website is following legal accessibility requirements of AODA or ACA

03 List the 4 guidelines and standards of web accessibility

Key Terms and Concepts

- ACA: Accessible Canada Act, an act to realize a barrier free Canada by 2040
- Accessibility: The process of reducing disability in products or websites
- AODA: Accessibility for Ontarians with Disabilities Act, an act to make Ontario fully accessible to people with disabilities by 2025
- ATAG: W3C Authoring Tool Accessibility Guidelines, authoring tools to make web accessibility more accessible to creators with disabilities.
- Disability: The act of not being able to do certain tasks based off of the environment
- Inclusion: The method of making things more usable and accessible. Broader than accessibility.
- Medical Model: An outdated model stating a person is disabled based off of their medical condition
- Social Model: Barriers made by society that limits a certain group of people creating a disability.

- UAAG: User Agent Accessibility Guidelines, guidelines for making web agents more accessible
- Usability: How intuitive a website is. How easy it is to use.
- UX: User experience, how a user perceives and navigate a website or application
- WAI-ARIA: WAI's Accessible Rich Internet Applications Guidelines, provides technical specifications for web applications to assistive technologies
- WCAG: Web Content Accessibility Guidelines, guidelines for creating website/content for people with visual, auditory, and cognitive impairments more accessible

Now that you have learned what a disability is and what standards and guidelines there are to improve web-based technology to remove disability barriers, you will next learn about assistive technology. In the next Module you will learn how Assistive technology is used to help people access website and navigating the web.

References

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Accessibility review reports using the WCAG assessment methodology

How to review your website

For this section, you will be focusing on the W3.org's WCAG Evaluation Methodology that can be found here: **https://www.w3.org/TR/WCAG-EM/ (https://www.w3.org/TR/WCAG-EM/)**

The first step in evaluating a website's accessibility is to determine its type and scope.
Is your website a small website with few pages, is it a web application or are there multiple different versions of the website (e.g. language or mobile versions)?
The differences in the type and scope determine the size of the evaluation.

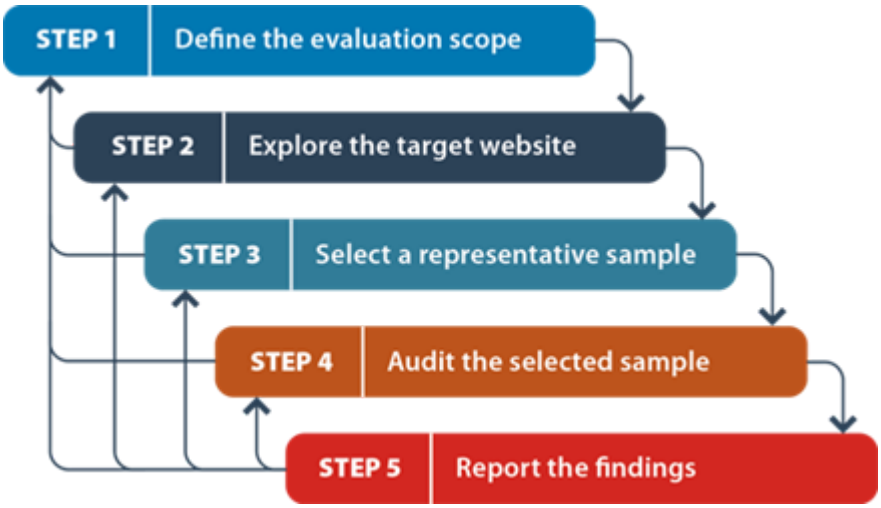
Next is determining who will be administering the evaluation. Are you performing it yourself, is there a third-party assessment being run on your site, is it being evaluated as it is being developed or will it be evaluated after it is finished, or finally will it be an aggregate of all the above?

After determining the context of the evaluation you move into the Evaluation Procedure stage.

In this stage you design the workflow of your evaluation

The following diagram shows five steps in sequence.

- Define the evaluation Scope
- Explore the target website
- Select a representative sample
- Audit the selected sample
- Report the findings



Define the evaluation Scope

In this step you plan the overall scope of your evaluation. What conformance level are you testing, what you are trying to achieve with this evaluation. If you are evaluating for someone else, like the owner of the website, then you will plan out this part with them to find their intent.

Explore the target website

Prior to your evaluation you need to view the website, to see its functionality, purpose and how to navigate or use it. Also determine the type of content you will be reviewing.
You want to figure out what the website is designed to do and where most traffic will be spent.

Find the most common webpages that will have the highest traffic and list them down as they will be used in the next step.

Select a Representative Sample

In this step you take your findings from the previous step and determine what needs testing.

There are many determining factors on what size of sample will be needed. How big is the website, the age of the website, consistency of the website, the development process of the site, if the site had a prior evaluation and how much has been changed since then.

You will want to take a selection of frequently used pages, home page, store, etc. then take a random 10% of other webpages. As part

Audit the Selected Sample

Now that you have your pages, the auditor will go through each one and evaluate them using the WCAG conformance requirements.

The Auditor will run both automated testing and manual testing on the website to ensure that the site works with AT, depending on the scope you may need to work with specialized individuals that use AT to interface with websites for stronger testing.

Report the Evaluation Findings

After completing the audit the auditor will create a report using their findings. This report covers the following:

- About the Evaluation
  - Name of the evaluator
  - Name of the evaluation commissioner
  - Date for the evaluation (completion date or duration period)

- Evaluation Scope
  - Scope of the website defined in Step 1.a: Define the Scope of the Website
  - Conformance target defined in Step 1.b. Define the Conformance Target
  - Accessibility support baseline defined in Step 1.c: Define an Accessibility Support Baseline
  - Additional requirements, if any, defined in Step 1.d: Define Additional Evaluation Requirements (Optional)
- Website Exploration
  - Web technologies relied upon identified in Step 2.d: Identify Web Technologies Relied Upon
  - Optional: Common web pages identified in Step 2.a: Identify Common Web Pages of the Website
  - Optional: Essential functionality identified in Step 2.b: Identify Essential Functionality of the Website
  - Optional: Variety of web page types identified in Step 2.c: Identify the Variety of Web Page Types
  - Optional: Other relevant web pages identified in Step 2.e: Identify Other Relevant Web Pages
- Representative Sample
  - Web pages selected through structured sampling in Step 3.a: Include a Structured Sample
  - Randomly selected web pages and selection method used in Step 3.b: Include a Randomly Selected Sample
  - Complete processes selected in Step 3.c: Include Complete Processes
- Sample Audited
  - Evaluation outcomes from Step 4.a: Check All Initial Web Pages
  - Evaluation outcomes from Step 4.b: Check All Complete Processes
  - Evaluation outcomes from Step 4.c: Compare Structured and Random Samples

“ (Website Accessibility Conformance Evaluation Methodology)

After the report is complete you will have the conformance levels of the website.

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Filtering the WCAG documentation, techniques & failures

What is WCAG?

WCAG stands for “Web Content Accessibility Guidelines”. They are a set of guidelines to improve web availability to user with disabilities trying to access them. Currently, they are in their version 2.1 state, however version 2.2 is in its draft form and awaiting approval.

Behind WCAG, there are four governing principles that websites should follow:

- Perceivable
- Operable
- Understandable
- Robust

This is another general-purpose page layout. Unlike the Basic page, this one includes samples of some commonly used page elements.

You can copy one, or more, of the specific elements from this page and paste them into other pages. By combining elements, you can create new layouts for your courses.

The following links are for the official WCAG documents:

WCAG version 2.1 (https://www.w3.org/TR/WCAG21/)

WCAG version 2.2 (Draft) (https://www.w3.org/TR/WCAG22/)

To navigate the document the Table of Contents Breaks it down into the following sections:

- Introduction
- Perceivable
- Operable
- Understandable
- Robust
- Conformance
- Glossary
- Input Purposes for User Interface Components

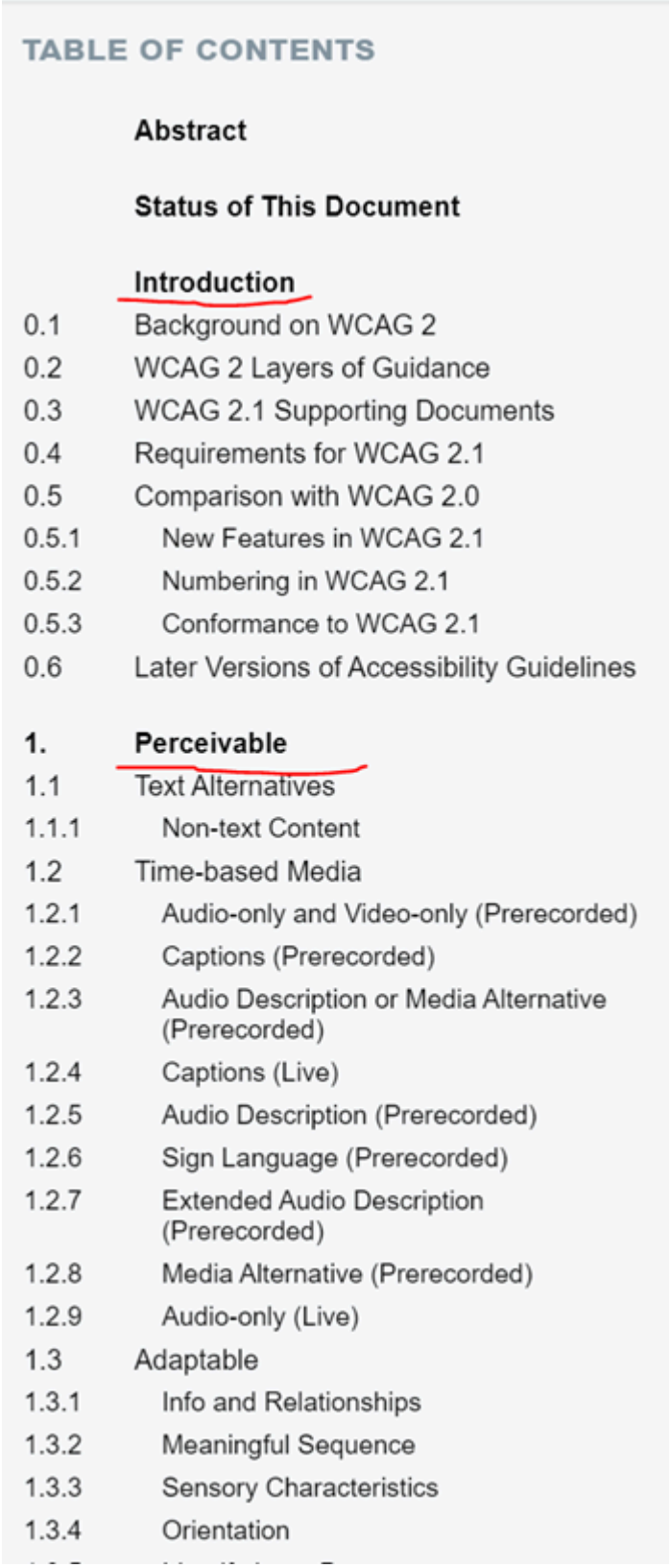


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Abstract

Status of This Document

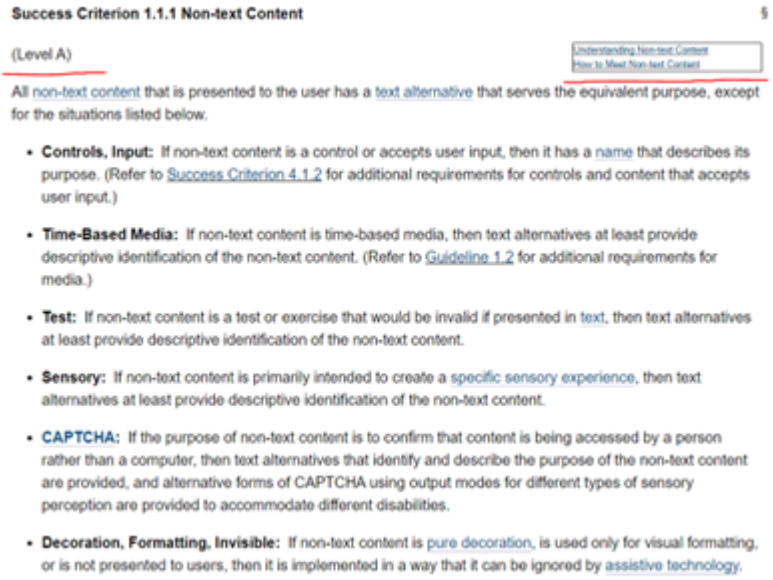
Introduction

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0.3 WCAG 2.1 Supporting Documents
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0.5 Comparison with WCAG 2.0
0.5.1 New Features in WCAG 2.1
0.5.2 Numbering in WCAG 2.1
0.5.3 Conformance to WCAG 2.1
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1. Perceivable

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For each subsection you can view what the guidelines are there, they also show you the compliance level for that guideline and how to implement it.



Success Criterion 1.1.1 Non-text Content §

(Level A)

Understanding Non-text Content
How to Meet Non-text Content

All non-text content that is presented to the user has a text alternative that serves the equivalent purpose, except for the situations listed below.

- **Controls, Input:** If non-text content is a control or accepts user input, then it has a name that describes its purpose. (Refer to Success Criterion 4.1.2 for additional requirements for controls and content that accepts user input.)
- **Time-Based Media:** If non-text content is time-based media, then text alternatives at least provide descriptive identification of the non-text content. (Refer to Guideline 1.2 for additional requirements for media.)
- **Test:** If non-text content is a test or exercise that would be invalid if presented in text, then text alternatives at least provide descriptive identification of the non-text content.
- **Sensory:** If non-text content is primarily intended to create a specific sensory experience, then text alternatives at least provide descriptive identification of the non-text content.
- **CAPTCHA:** If the purpose of non-text content is to confirm that content is being accessed by a person rather than a computer, then text alternatives that identify and describe the purpose of the non-text content are provided, and alternative forms of CAPTCHA using output modes for different types of sensory perception are provided to accommodate different disabilities.
- **Decoration, Formatting, Invisible:** If non-text content is pure decoration, is used only for visual formatting, or is not presented to users, then it is implemented in a way that it can be ignored by assistive technology.

Finally the WCAG document has a section on techniques to implement some of the guidelines. These are excellent documents that give you instructions and code examples to one how those guidelines are met.

For WCAG 2.1 techniques use the following link:
https://www.w3.org/WAI/WCAG21/Techniques/ (https://www.w3.org/WAI/WCAG21/Techniques/)

The document also shows common failures to the guidelines and how to remedy them.
https://www.w3.org/TR/WCAG20-TECHS/failures.html (https://www.w3.org/TR/WCAG20-TECHS/failures.html)

An example of one of the failures is as follows:
“F1: Failure of Success Criterion 1.3.2 due to changing the meaning of content by positioning information with CSS” (Failures)

This failure uses CSS to build a table. The table has two columns. Products and Locations with three items in the Products table and two in the Locations table.

Fig. 3.3 shows the HTML and what the items look like before CSS is applied.

HTML

```
<div class="box">
    <span class="menu1">Products</span>
    <span class="menu2">Locations</span>
    <span class="item1">Telephones</span>
    <span class="item2">Computers</span>
    <span class="item3">Portable MP3 Players</span>
    <span class="item5">Wisconsin</span>
    <span class="item4">Idaho</span>
</div>
```

Products Locations Telephones Computers Portable MP3 Players Wisconsin Idaho

Fig. 3.4 Shows the CSS added to the code.

HTML

CSS

```
.menu1 {
    position: absolute;
    top: 3em;
    left: 0em;
    margin: 0px;
    font-family: sans-serif;
    font-size: 120%;
    color: red;
    background-color: white
}
.menu2 {
    position: absolute;
    top: 3em;
    left: 10em;
    margin: 0px;
```

Products Locations

Telephones
Computers
Portable MP3 Players

Idaho
Wisconsin

As you can see, to a person who can view the page with their eyes, it will look fine. However, this will cause a big problem with screen readers as the screen reader cannot use the CSS to differentiate the objects in the list. The Screen reader would read it out as if it did not have the positioning established e.g. "Products Locations Telephones Computers Portable MP3 Players Wisconsin Idaho."

The suggested solution to this is to use either more meaningful elements like a table, or a definition list.

With the prevalence of CSS and JS used for layout and web design the more important it is to build an accessible web page.

To summarize WCAG please watch the following video: **https://www.youtube.com/watch?v=5H1JGdqLrWo (https://www.youtube.com/watch?v=5H1JGdqLrWo)**

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Accessibility Guidelines (WCAG)

In this module you will learn what WCAG is and how its guidelines and principles are used to test a website's accessibility level. You will also learn how an accessibility audit is made and how it determines a website's accessibility compliance.

Learning Objectives

By the end of this module, you should be able to complete the following:

01 List the four principles of WCAG

02 Use the level 3 levels of conformance to determine a website's accessibility level

03 Apply the WCAG evaluation methodology to create an accessibility audit for a website

Key Terms and Concepts

1. Accessibility Audit: A number of tests and checks to determine how well a website is for accessibility
2. AT: Assistive Technology
3. Compliance Levels: A rank of how accessible a website is to someone with disabilities
4. CSS: Cascading Style Sheets. Code that changes the appearance of HTML Code
5. HTML: Hypertext Markup Language: A file that constructs a website using tags and structures
6. Operable: Any user may be able to use a website
7. Perceivable: A website is able to be viewable/interactable to anyone
8. Robust: A website can be compatible with any device or AT hardware
9. Time-Based Media: Any images, video, or time-based interactive elements on a website.

10. Understandable: Content and layout makes sense and can be understood by anyone

11. WCAG: Web Content Accessibility Guidelines

Use of WCAG in acquisition of software, hardware, electronic content, support documentation and services

You may find during your web and cloud development journey that not all the software that you will use will be developed by you. Many websites and software use either open source or proprietary libraries and modules in them, since it is usually cheaper to license or credit other companies who have made something than to develop it from scratch.

The issue here comes when you are using that software and you cannot comply with WCAG standards.

Some examples of these could be things like online maps or complex diagrams that prevent people with visual disabilities to be incapable of accessing them. It could also be that the modules or software you are using may be outdated and from an older standard of WCAG.

If this is the case and the software you are using is Open source, you may have to ask for updates to be WCAG compliant or fork a branch of the repository and update it yourself.

The trouble comes from proprietary software that is locked under a legal license. You may have to talk to the Licenser to update the code.

If none of these solutions can be met, you may have to have a summary of unconvertable content on your site.

When acquiring software for your applications or sites, ensure they are WCAG compliant when possible. Finally, if you have any documentation or files on your site, ensure that they too are compliant and AT friendly.

Remember, accessibility is essential for some, useful for all.

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WCAG audit process

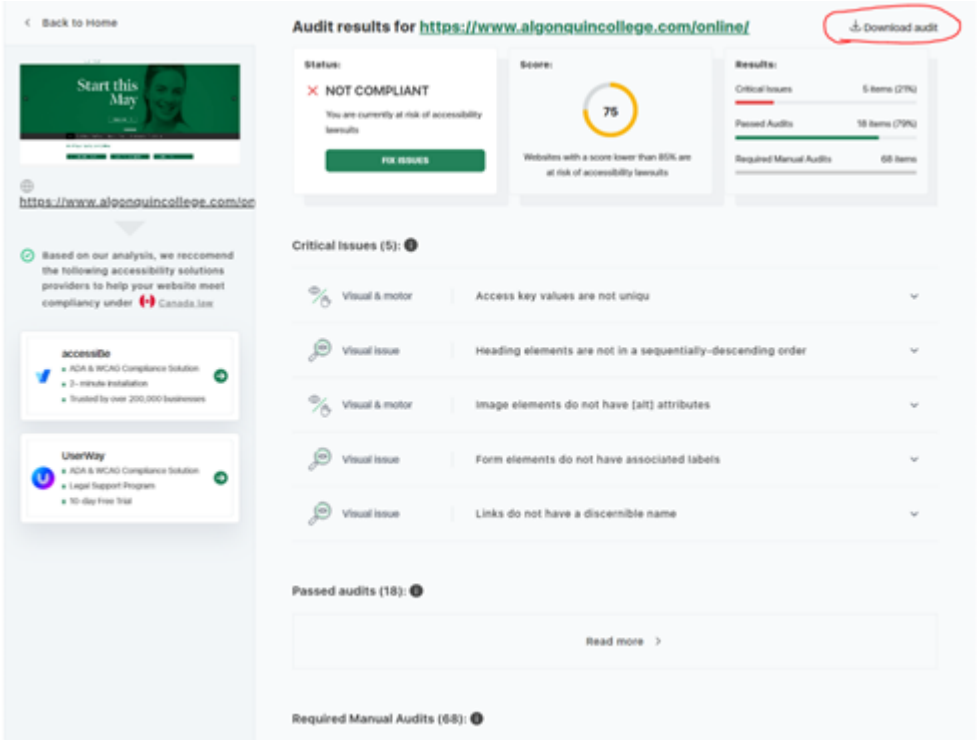
How do you tell if your website is accessible?
There are many online tools for checking accessibility of a webpage. Many companies will charge an Audit fee to look over your complete website and send a report back to you with guidelines you have met and outstanding guidelines that are preventing you from getting a conformance level.

Now these web services are usually used for large scale websites with numerous pages. If you are developing a small website and want to test your accessibility performance, how can you do it?

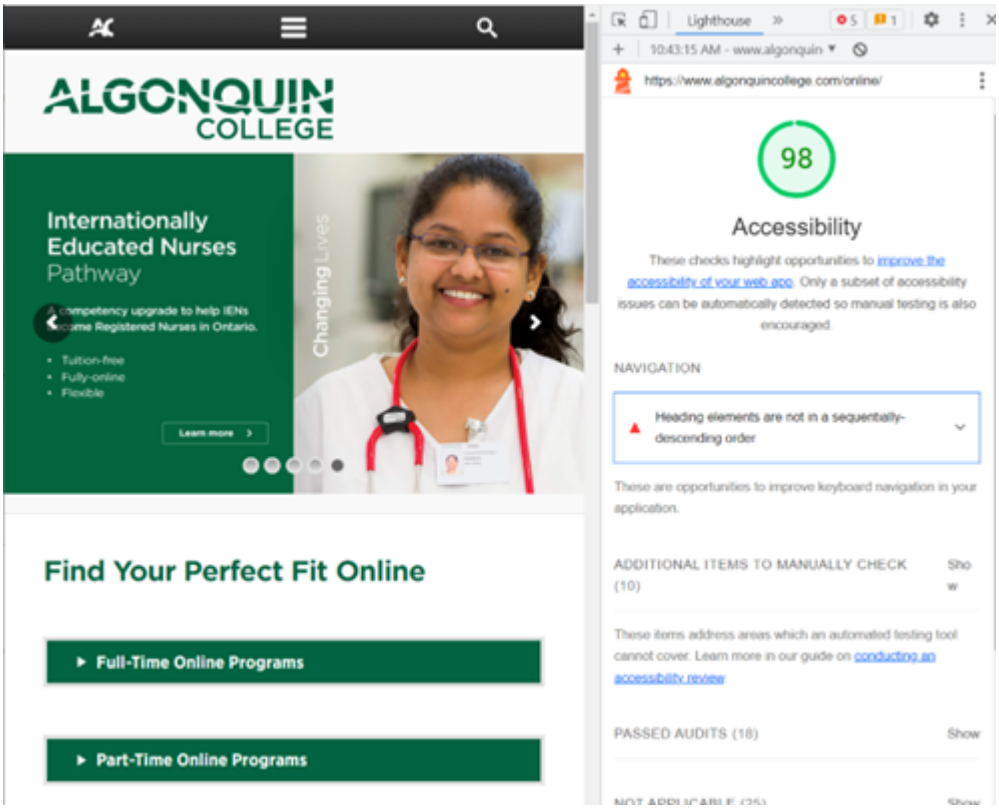
One free Accessibility checker can be found here: **https://www.accessibilitychecker.org/ (https://www.accessibilitychecker.org/)**

To perform it you need to paste a URL's website into the search criteria then choose the country you want to be compliant in.

In the following example Algonquin College's website is being used.



In the following example Algonquin College's website is being used.



The Audit will take a few minutes to run, then will display the results on a new page.

It will give you a Boolean status of compliant or not compliant, a score based off of what tests you have passed and results of your audit.

It then lists all the issues that are critical in having an accessible site.

Finally, this is just a small, automated audit. There are more manual tests that need to be tested and addressed. These are tests like, keyboard navigation and how it works on screen readers to name a couple.

This Audit also allows you to print a single time report if needed.

While developing you can use built in browser tools (like Lighthouse in Chrome) to give you a quick audit into your site's current accessibility.

While this gives you a quick look and will pick up simple errors there is always more to test.

In the next section you will learn about the assessment methodology on how an accessibility audit will be applied to a website.

For a recap on accessibility audits please watch the following video **(https://www.youtube.com/watch?v=LvU-Ie-ilb4)**

- **https://www.youtube.com/watch?v=LvU-Ie-ilb4 (https://www.youtube.com/watch?v=LvU-Ie-ilb4)**

Finally, more testing will be covered in the next module.

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WCAG principles, guidelines, success criteria and levels of compliance.

Levels of Compliance

Before you learn about the guidelines you should first learn about compliance levels.

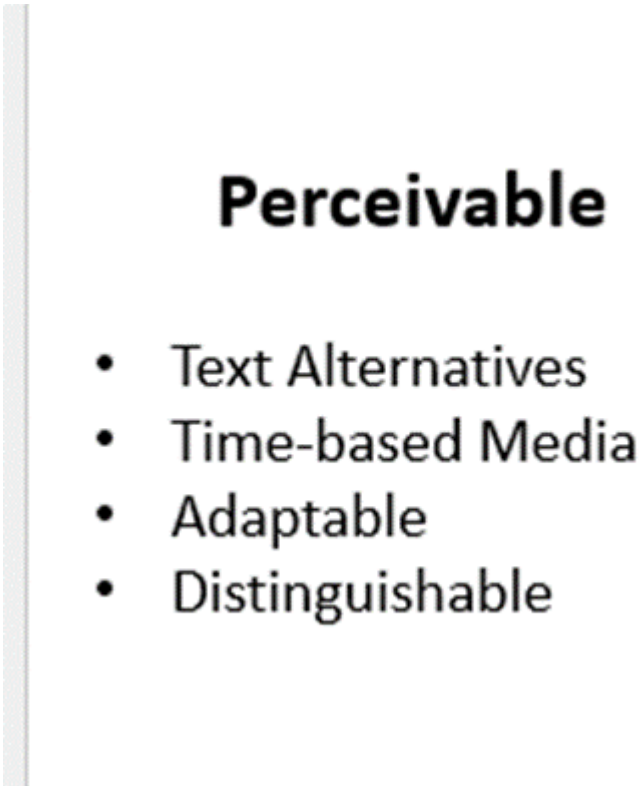
For each guideline there will be a compliance level between A (Lowest), AA or AAA (Highest). If your website meets at least all of the level A guidelines, then your site can be considered to have Level A conformance. A website with all Level AA and level A guidelines can be considered Level AA conformant. This is where MOST big businesses and Government websites try to achieve.

Finally if you have all three conformance level guidelines then you are considered Level AAA conformant.

Level A is the basic level that is easy to achieve and has a minimal impact on the website.

All guidelines should be easily testable to verify whether your site is compliant or not.

Perceivable



Perceivable means that components and information on a website should be presentable to all types of users.

Guideline 1.1 Text Alternatives. (Level A)

Non-text content directed towards users should have a text alternative that is equivalent in purpose. This is like an Alt text on an image or captcha.

Guideline 1.2 Time-Based Media

All time-based media is content that moves, makes noise, displays synchronized content or content that changes without user input. It can be video only, audio only or a combination of both.

The guidelines have nine sub guidelines of various conformance levels.
It is mostly suggesting alternatives for any Time-based media.

Such as:

- Audio and Video (Pre-recorded) [A]
- Captions (Pre-recorded Media) [A]
- Audio Description or Media Alternatives (Pre-recorded) [A]
- Captions (Live Media) [AA]
- Audio Description (Pre-recorded) [AA]
- Sign Language (Pre-recorded) [AAA]
- Extended Audio Description (Pre-recorded) [AAA]
- Media Alternative (Pre-recorded) [AAA]
- Audio-Only (Live) [AAA]

Guideline 1.3 Adaptable

The Content on your website is able to be viewed in diverse ways. Usually setting out a simpler layout without a loss of information or structure.

An example is the structure of a website viewed by a desktop computer would not work well on a mobile device. Having a site that can adapt based on the environment it is viewed on is a great benefit.

Having adjustable text size and colour done through CSS also improves use for people with low vision or colour blindness.

This guideline is broken down into six sub-guidelines:

- Info and Relationships [A]
- Meaningful Sequence [A]

- Sensory Characteristics [A]
- Orientation [AA]
- Identify Input Purpose [AA]
- Identify Purpose [AAA]

Guideline 1.4 Distinguishable

This guideline makes it easier for users to retrieve information from a website by sight or hearing. It also allows for differentiation between foreground and background elements.

This guideline is broken down into thirteen sub-guidelines:

Use of Color [A]

Audio Control [A]

Contrast (Minimum) [AA]

Resize text [AA]

Images of Text [AA]

Contrast (Enhanced) [AAA]

Low or No Background Audio [AAA]

Visual Presentation [AAA]

Images of Text (No Exception) [AAA]

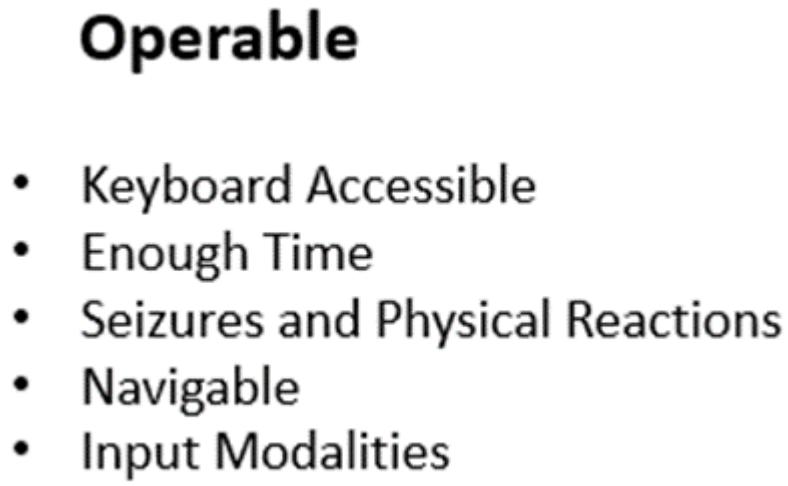
Reflow [AA]

Non-text Contrast [AA]

Text Spacing [AA]

Content on Hover or Focus [AA]

Operable



The over arching goal of the Operable principle is to allow users access to navigate and interact with your website. It is broken down into the five following guidelines:

Guideline 2.1 Keyboard Accessible

This guidelines name says it all, it allows users to interact with a website using their keyboard only.

This guideline is broken down into four sub-guidelines:

- Keyboard [A]
- No Keyboard Trap [A]
- Keyboard (No Exception) [AAA]
- Character Key Shortcuts [A]

Guideline 2.2 Enough Time

This guideline states that a user should have enough time to read and interact with content on a webpage.

This guideline is broken down into six sub-guidelines:

- Timing Adjustable [A]
- Pause, Stop, Hide [A]
- No Timing [AAA]
- Interruptions [AAA]
- Re-authenticating [AAA]

- Timeouts [AAA]

Guideline 2.3 Seizures and Physical Reactions

Ensure your site will not cause seizures or physical reactions for users.
At a minimum, if your site does have flashing content, ensure it is within the recommended bounds of the first sub-guideline.

This guideline is broken down into three sub-guidelines:

- Three Flashes or Below Threshold [A]
- Three Flashes [AAA]
- Animation from Interactions [AAA]

Guideline 2.4 Navigable

Users can navigate a webpage, find content, and can determine where they are. This does not just help users with disabilities, but everyone. It is really hard to navigate a poorly designed webpage.

This guideline is broken down into ten sub-guidelines:

- Bypass Blocks [A]
- Page Titled [A]
- Focus Order [A]
- Link Purpose (In Context) [A]
- Multiple Ways [AA]
- Headings and Labels [AA]
- Focus Visible [AA]
- Location [AAA]
- Link Purpose (Link Only) [AAA]
- Section Headings [AAA]

Guideline 2.5 Input Modalities

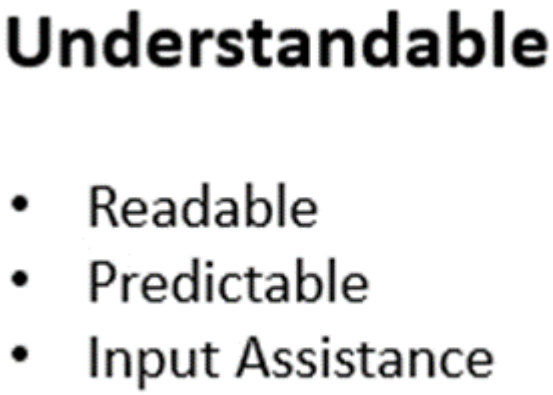
Users are able to input and operate a website through multiple different inputs such as Gestures or touch screens to name a few.

This guideline is broken down into six sub-guidelines:

- Pointer Gestures [A]

- Pointer Cancellation [A]
- Label in Name [A]
- Motion Actuation [A]
- Target Size [AAA]
- Concurrent Input Mechanisms [AAA]

Understandable



This principle states that the webpage's information and how it operates can be understood. This can be setting a language of a website to be interpreted by an AT (assistive technology) reader in the correct language, for example.

There are 3 Guidelines in this principle.

Guideline 3.1 Readable

Users are able to read and understand the content on the page. "Read" in this case can be interpreted as visual reading or AT reading such as a narrator or braille device.

To get a feel for how frustrating it could be, imagine reading a technical textbook where they never explain what the abbreviations meant/stood for. You may understand some of it but there is a lot of information that is lost, and it would not be a very pleasant read.

This guideline is broken down into six sub-guidelines:

Language of Page [A]

Language of Parts [AA]

Unusual Words [AAA]

Abbreviations [AAA]

Reading Level [AAA]

Pronunciation [AAA]

Guideline 3.2 Predictable

A webpage should act and look in a normal way that would lead to predictable navigation and function.

This guideline is broken down into five sub-guidelines:

- On Focus [A]
- On Input [A]
- Consistent Navigation [AA]
- Consistent Identification [AA]
- Change on Request [AAA]

Guideline 3.3 Input Assistance

What this guideline looks for is that a webpage has some QoL (quality of life) features to improve user input. It starts with simple error identification (incorrect password, wrong input etc.) and leads to error prevention (Reversable submissions and validated data).

This guideline is broken down into six sub-guidelines:

Error Identification [A]

Labels or Instructions [A]

Error Suggestion [AA]

Error Prevention (Legal, Financial, Data) [AA]

Help [AAA]

Error Prevention (All) [AAA]

Robust



The Final principle is Robust. What this principle is looking for is that webpages and applications are able to be read and interpreted by many different user agents, such as assistive technology.

Guideline 4.1 Compatible

This guideline wants your webpage to have as much compatibility with current and future user agents.

This guideline is broken down into three sub-guidelines:

- Parsing [A]
- Name, Role, Value [A]
- Status Messages [AA]

While this section gives you a top-level overlook into each principle and guideline it is highly encouraged to read the guidelines individually for a more in-depth explanation into each one's success criteria.

That was a lot of reading, watch the following video for a quick visual recap on what WCAG is, what the conformance levels are and what its principles are.

- **https://www.youtube.com/watch?v=FsGussrOX60 (https://www.youtube.com/watch?v=FsGussrOX60)**

Finally, there are **9 new Success Criteria in the WCAG 2.2's draft. (https://www.w3.org/TR/WCAG22/#comparison-with-wcag-2-1)**

They are:

- 2.4.11 Focus Appearance (AA)
- 2.4.12 Focus Not Obscured (AA)
- 2.4.13 Focus Not Obscured (Enhanced) (AAA)
- 2.5.7 Dragging Movements (AA)
- 2.5.8 Target Size (Minimum) (AA)
- 3.2.6 Consistent Help (A)
- 3.3.7 Accessible Authentication (AA)
- 3.3.8 Accessible Authentication (Enhanced) (AAA)
- 3.3.9 Redundant Entry (A)

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